

Sunland Adventure Park Software

Requirement Specification

Digital Dreams

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Table of Contents

Sunland Adventure Park Software Requirement Specification	1
1. Customer Statement of Requirements.....	3
2. Glossary Of Terms	6
3. Functional Requirements Specification.....	7
3.1 Stakeholders	7
3.2 Actors and Goals	7
3.3 Use Cases.....	8
Figure 3.3.1 Use Case Diagram of System.....	8
Figure 3.3.2 Use Case Description for Adding a Family Member	9
Figure 3.3.3 Use Case Description for Making a Reservation	11
Figure 3.3.4 List of Casual Descriptions	13
3.4 Activity Diagrams	14
Figure 3.4.1 Adding a Family Member Activity Diagram	15
Figure 3.4.2 Opening Interactive Map Activity Diagram.....	16
3.5 Interaction Diagrams	17
Fig 3.5.1 Sequence Diagram for Buying a Ticket.....	17
Figure 3.5.2 App Behavioural State Machine Diagram	19
3.6 Overview Class Diagram	20
Figure 3.6.1 Overview Class Diagram:	20
4. Nonfunctional Requirements	21
5. References	23
6. User Interface	23
Figure 6.1 Interactive Map UI	24
Figure 6.2 Profile Page UI	25
Figure 6.3 Home Page UI	26
Figure 6.4 Ticket Page UI	27
7. Appendix.....	28
8. Effort Breakdown Table.....	28
Figure 8.1 Effort Breakdown Table:	29

1. Customer Statement of Requirements

Sunland Adventure Park is requesting an app be made for their amusement park. The park is hopeful that the implementation of the app will boost business by allowing customers easier access to information, tickets, reservations, activities, shops, and more. The app will include an interactive map that allows guests to see where they are in the park. The map will also show the locations of family members, and when permissible, friends.

The app will be friendly for all ages. It will be easy to use and will require no training for guests to understand the functionality of the app. We'll be implementing a Help button in case a guest needs help using the app. Employees will need some

training in order to become familiar with the app. The park's IT support will need a moderate amount of training in order to conduct maintenance updates on the app, though they are always welcome to reach out to our team and we will provide support. Overall, the app will provide a happier experience for guests at the park and it will provide the park an opportunity to bring in more revenue.

Here is the cost-benefit analysis for this app:

Assumptions:

1. Dollar value increases by 3.5 % per year.
2. Labor costs are estimated at \$85 per man-hour.

Estimated Costs:

1. Equipment Costs: Assume \$50,000 for necessary equipment(Source: Industry Average).
2. Network Costs: Assume \$50,000 for setting up and maintaining the network
3. Software Costs: Assume \$75,000 for licenses and implementation
4. Labor costs: Assume 1,000 hours per year at \$85 per man-hour, totaling \$102,000 per year.

Cost-Benefit Analysis:

Benefits: Increased Sales, 10% more visitors, 404 extra people a day, 102.23 made off each guest per day

Subscription: access to fast pass, assume 3777 made per year.

Below is a list of requirements that the system should be able to do.

1. The app must handle reservations.
2. The app must handle transactions.
3. The app must include an interactive map.

4. The app must automatically connect and disconnect with the park when the user is entering and leaving the park.
5. The app must include a guest portal.
6. The interactive map must show the location of the user and their family.
 - a. The map will also include the option for users to share their location with friends.
7. The map will show the location of nearby activities, shows, shops, rides, and restaurants available to the guest.

2. Glossary Of Terms

The glossary of terms contains words that may be unknown to the client or person(s) reading this document. The words included in the glossary are used in this document. The word is listed, followed by the definition for the word.

1. Actors: The people, systems, or objects directly interacting with the proposed system. E.g. employees, guests, management, database system.
2. Ping/Pinging: To signal to the selected device querying about its network location.
3. Multiplicities: How many of one object can be associated with how many of another object.

3. Functional Requirements Specification

3.1 Stakeholders

The stakeholder's section identifies all persons of interest regarding the requested system. All stakeholders for the Sunland Adventure Park system are listed below:

Park Manager

Paying Customers

Restaurant Employees

Development Team

Ride Operators

3.2 Actors and Goals

This section identifies who interacts directly with the system. There are two types of actors that interact with the system. The first is an initiating type, and the second is a participating type. Actors will be listed below along with their type. The actors will also have goals listed with them.

Initiating:

Development Team (Us) - Making the app work with the system as well as having all the correct functionalities.

Park Manager - Maintains system as well as making sure it actually implements

correctly, they do this by testing it against their current network and letting us know what works and what doesn't.

Participating:

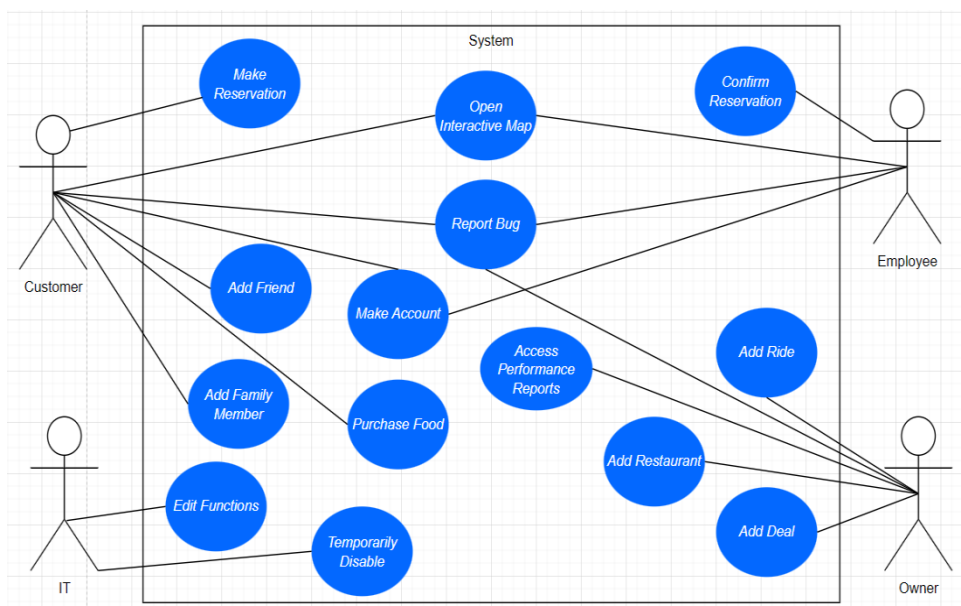
Ride Operators - Use the app to better save time and money by not having to direct customers as much as well as making the ticket process easier.

Customers - Use the app to better their experience at the park, use it to reserve tickets, find friends, eat out, guide them, etc...

3.3 Use Cases

This section identifies use cases. A use case is a description of the ways in which a user interacts with a system. Use cases help outline the ways in which a user will interact and use the system.

Figure 3.3.1 Use Case Diagram of System



This is a diagram that displays all the different new Use Cases added by and contained within the System. Lines indicate that the specific use case can be used by the specific individual.

Figure 3.3.2 Use Case Description for Adding a Family Member

Use case name: Adding Family Member	ID: MR	Importance Level: 3 of 5
Primary actor: Guest	Use case type: System	
Stakeholders and interests: Guests, Managers, Employees, and the Owner all take interest in this action. The employees, managers, and owners are counting on this system to add family members and friends to help boost guest satisfaction.		
Brief description: The guest will log into the app and head to the account page. They will then click “Add Family Member”. They can input all the information needed to add a family member, and then the member will then show up on the interactive map when within park grounds.		
Trigger: Whenever the guest taps on the “Add Family Member” button.		

Relationships:

Association: Guests

Include:

Extend:

Generalization:

Normal flow of events:

1. Guest will log into their account.
2. Guest will navigate to account page
3. Guest will tap on "Add Family Member" button
4. Guest will input all required information to add family member
5. Guest will click "Submit"
6. Added members will now show up on the interactive map.

Subflows:

1. The guest has the option to add multiple family members at a time.
2. System will automatically add the member to the interactive map once the guest hits "Submit"

Alternate/exceptional flows:

1. The guest has the option to cancel out of adding a family member through a cancel button.

This is the use case description for adding a family member. It shows us how the guest will interact with the app and the steps required in order to add a family member to their account.

Figure 3.3.3 Use Case Description for Making a Reservation

Use case name: Open Interactive Map	ID: MR	Importance Level: 4 of 5
Primary actor: Guest	Use case type: System	
Stakeholders and interests: Guests, Managers, and Owners have a high interest in this use case. The interactive map allows guests to see family members' locations and the locations of attractions, shops, restaurants, and shows within the park. It is a great tool for the guests as well as advertising the different events going on.		
Brief description: The guest will log into the app, and head to the map page. The interactive map will automatically be displayed. Guests do have the option to view the map when not in parkgrounds but it will not be interactive.		
Trigger: The guest taps on the map icon at the bottom of the screen or on the home page.		

Relationships:

Association: Guest

Include:

Extend:

Generalization:

Normal flow of events:

1. Guest will log into the app
2. Guest will tap on the map icon
3. The interactive map will display
4. Guest can interact with map

Subflows:**Alternate/exceptional flows:**

1. When not within park grounds, the map will not be interactive, but it will still display
2. Guest has the option of tapping on the Map button on the home page as well as the map icon.

This is the description for opening the interactive map. It will show us the steps that guests will need to take in order to open the interactive map. It also shows us if there are any alternate ways to get the map open.

Figure 3.3.4 List of Casual Descriptions

Below is a list of Casual descriptions for each and every use case added by the System.

- **Make Reservation:**
The user can make a reservation through the app that includes time, payment, and attraction.
- **Open Interactive Map:**
Anyone on the app can very easily open the interactive map which shows your location relative to different attractions and whoever you've friended on the system!
- **Confirm Reservation:**
Employees can use the app to confirm whether or not a user has a reservation just by looking at their screen, inputting an employee code, and hitting confirm, doing this grants the user access to the ride.
- **Report Bug:**
This button is accessible to anybody who uses the app, it will allow the user to input any bugs that they find in a chat box that will go to both the owner and IT.
- **Add Friend:**
This use case allows any user with an account to send a friend request to another user, if 2 users have requests both sent to each other the app will declare them friends and they will now show up on each other's maps. Tapping on a friend's location icon on the map opens up a chat where you can communicate with the other user.
- **Make Account:**
Customers and Employees are able to make accounts, this includes a username, password, picture, as well as a spot to input a special ID allowing employees to get access to different options and giving users with a seasonal pass access to their benefits.
- **Add Ride:**
This is only available to the owner and simply adds a ride onto the interactive map, with pricing, pictures, a description, and a name.
- **Add Deal:**
On the front of the home page deals will be shown along with different rides that the user has access to. Adding a deal can only be done by the owner and allows it to be advertised to the users when they open their app.
- **Add Restaurant:**
Same idea as adding a ride. Allows a user to add a restaurant onto the interactive map, pricing details, pictures, descriptions, and names.
- **Purchase Food:**

The user will be able to purchase food through the app which will send the money straight to the food workers and will be available to pick up and eat at the various restaurants.

- **Add Family Member:**

Any account can have any amount of family members, adding a family member makes it so that their name appears red on the interactive map so it is easier to find them in case you have a bunch of friends that will also appear.

- **Edit Functions:**

This is only available to IT and allows IT to add extra functions if necessary for later.

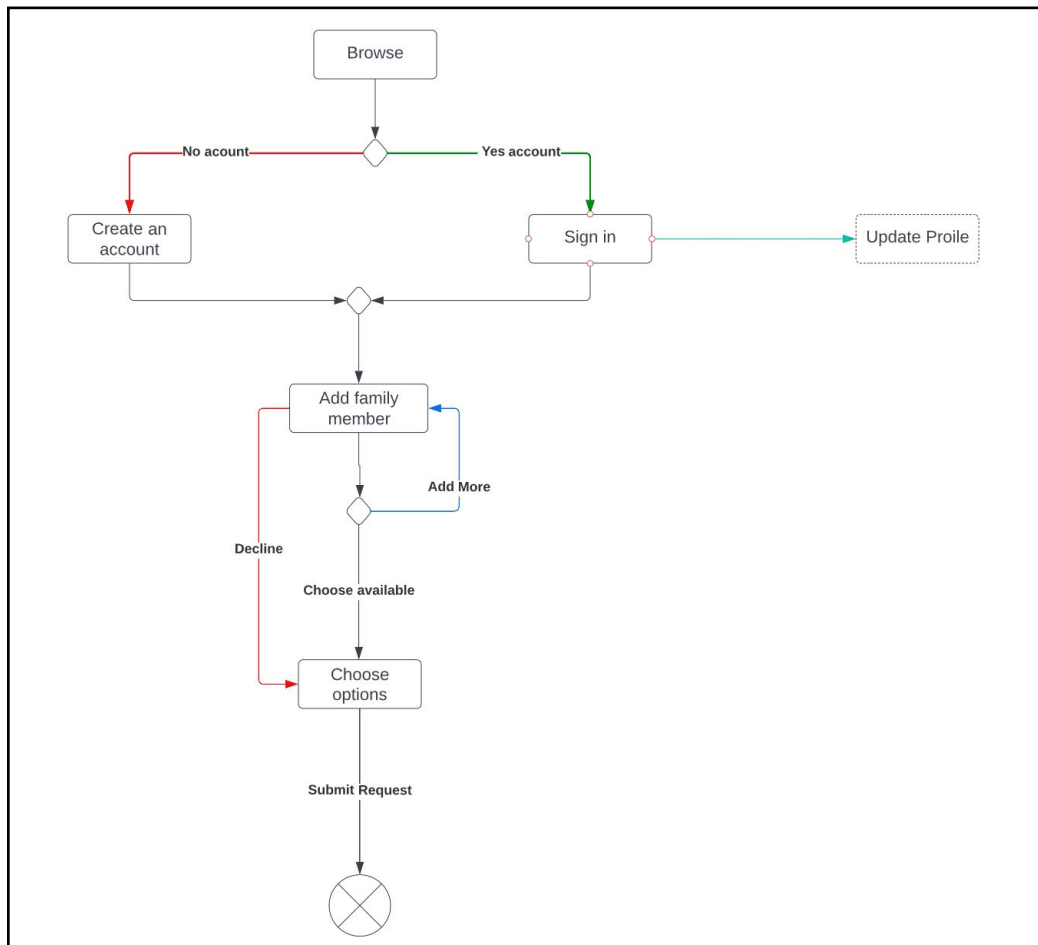
- **Temporarily Disable:**

This is available only to IT and can be activated if the service is under attack or a disablement is necessary.

3.4 Activity Diagrams

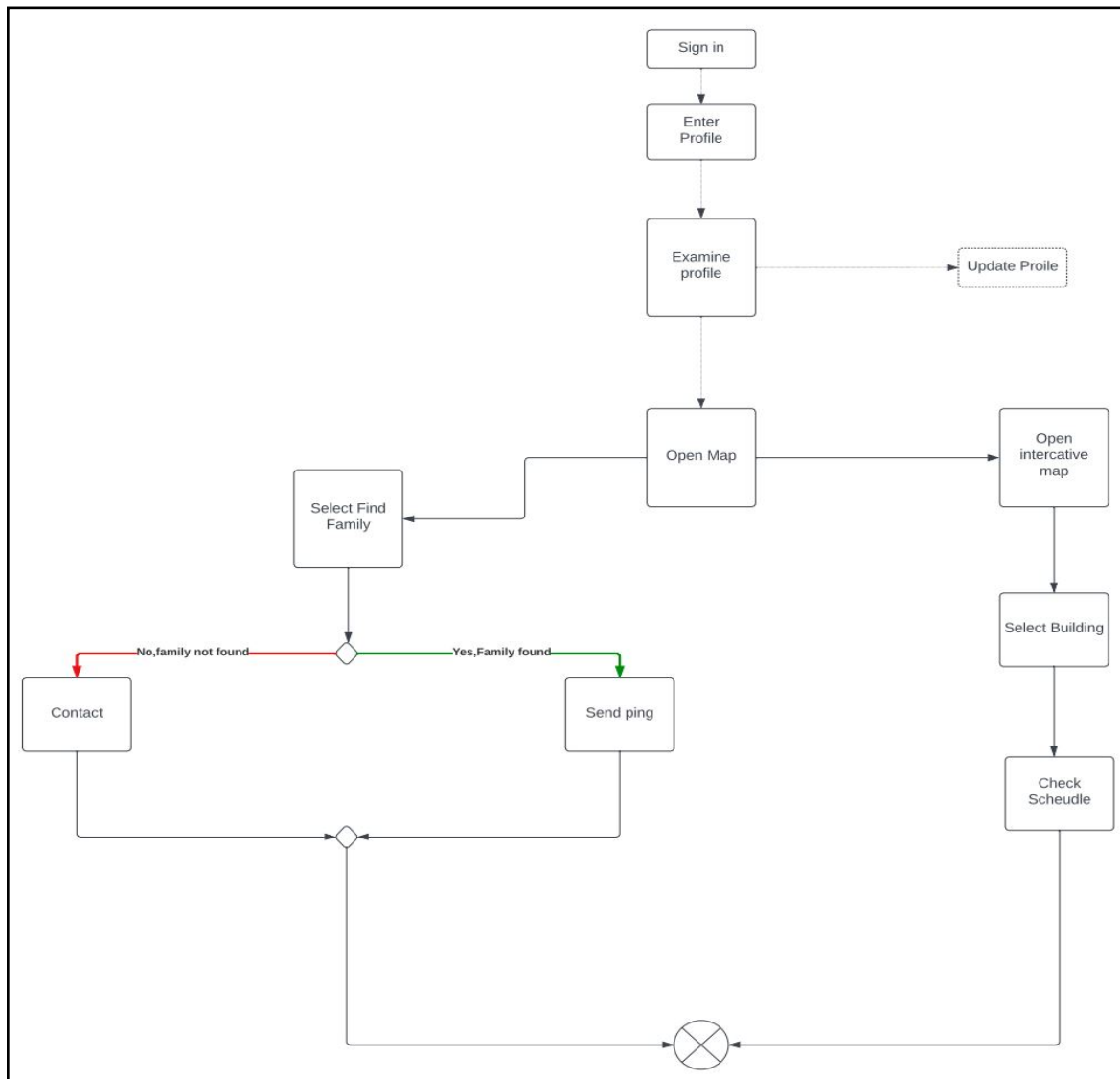
This section displays the activity diagrams for the proposed system. Activity diagrams are visual representations that show the steps and actions for a particular activity. It shows the client the set of steps an actor will need to take to complete an activity.

Figure 3.4.1 Adding a Family Member Activity Diagram



This is an activity diagram for adding a family member to your account. This diagram shows the actions and what they all lead to in the process of adding a family member.

Figure 3.4.2 Opening Interactive Map Activity Diagram

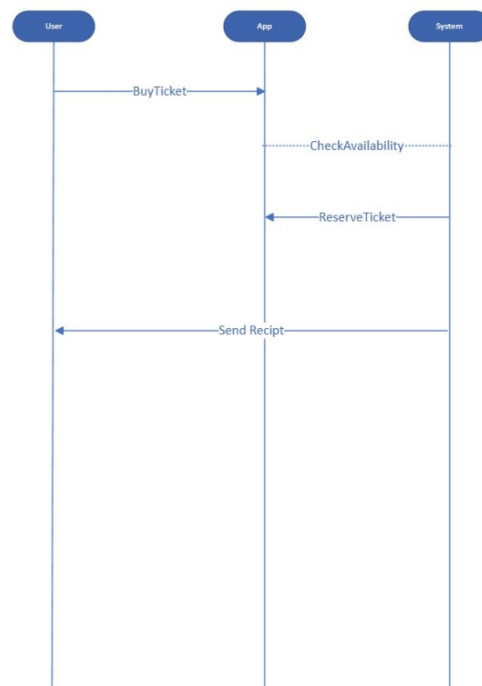


This is the activity diagram for the interactive map. Specifically, this diagram shows how to open the interactive map, find family members through a pinging system, contact family members through the app (if the app cannot locate the family member through a ping), and select and view a merchant/attraction schedule.

3.5 Interaction Diagrams

Interaction Diagrams show how objects within the system communicate and connect with each other. There are two kinds of interaction diagrams. The first is the communication diagram, and the second is the sequence diagram. The communication diagram shows how different objects within the system communicate while the sequence diagram focuses on the sequence of events the system takes during communication.

Fig 3.5.1 Sequence Diagram for Buying a Ticket

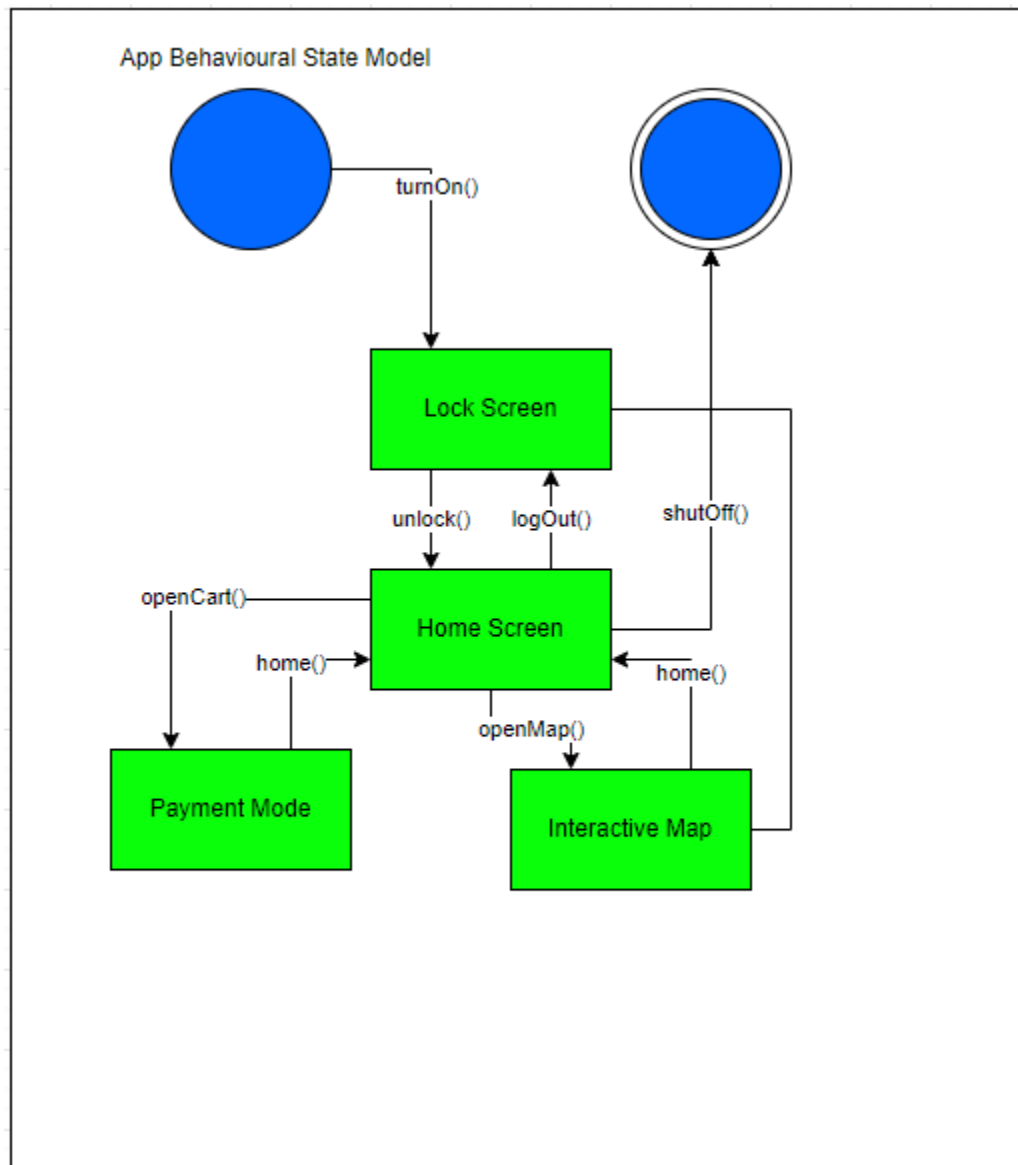


Below is a list of objects that all change states during their lifetimes.

- System
- Network
- Website
- App

We've also included a behavioral state machine diagram for the App object.

Figure 3.5.2 App Behavioural State Machine Diagram

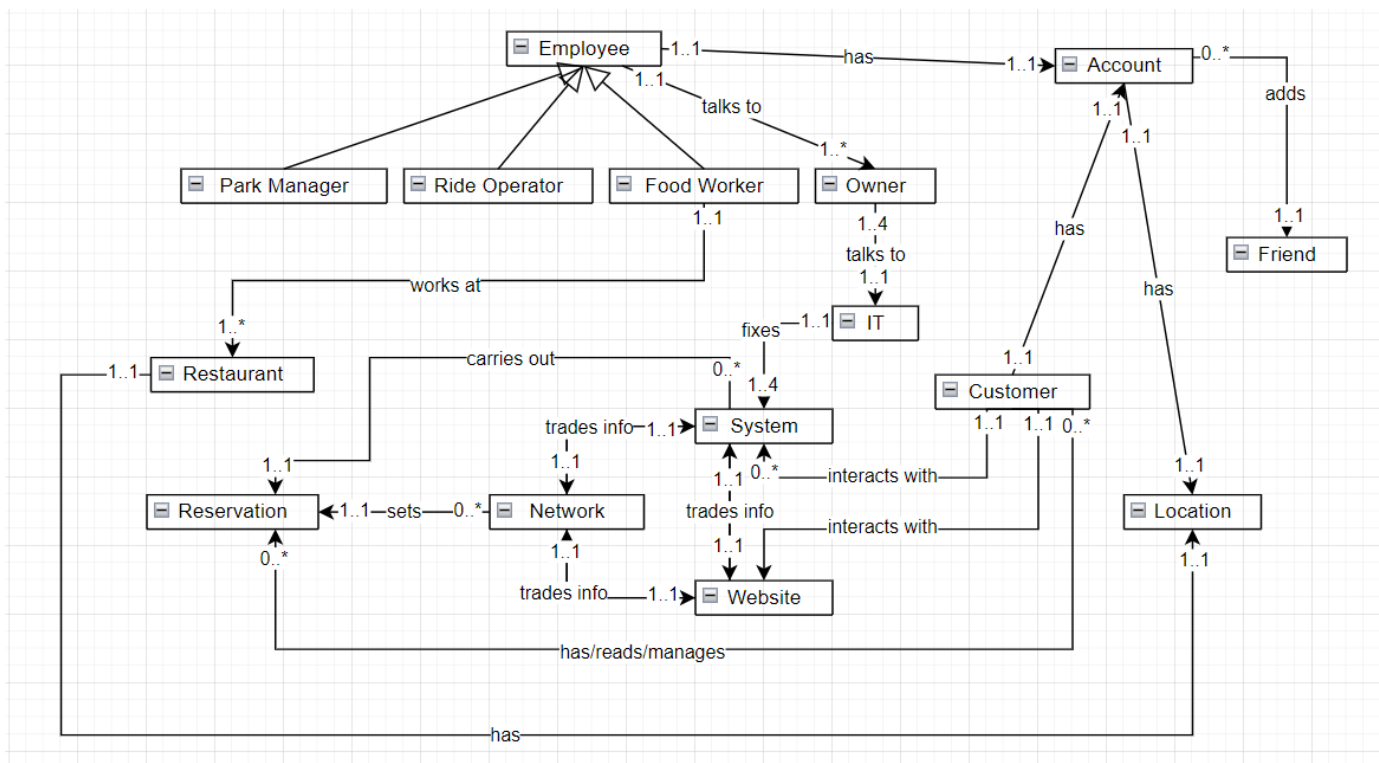


This is a model that shows how the app changes its behavior from off to on and the several stages it goes through to get there. Lock Screen is the only screen available after turning on the app then the home screen or off, from the home screen, you can go to paying or to the map.

3.6 Overview Class Diagram

This section identifies each of the classes within the system and the relationships between each class. Below is the overview diagram for all the classes and their relationships.

Figure 3.6.1 Overview Class Diagram:



The overview class diagram shows how every object of interest interacts with the system and each other. Has numbers to indicate **multiplicities** and words to display how they interact in general.

Below is a list of potential class domains (groups of objects) associated with the system.

1. Software Domain: Website, System, Network, Account, Reservation, Friend
2. Worker Domain: Employee, Ride Operator, Food Worker, Park Manager, IT, Owner
3. Systems Operator Domain: IT, Owner
4. Users Domain: Employee, Customer
5. Food Domain: Restaurant, Food Worker, Customer

4. Nonfunctional Requirements

Performance: The app will be able to handle a large number of users simultaneously without slowing down or crashing. Response time for requests will be quick and efficient.

Usability: The app will be easy to use and navigate, with clear and concise instructions for users. It should have a user-friendly interface with an intuitive design and appropriate colors and font sizes.

Reliability: The app will be always dependable and available for use, with minimal downtime or disruptions. It will have a backup and disaster recovery plan in case of system failures or crashes.

Security: The app will be secure and protect user data, personal information, and payment details from unauthorized access or cyber-attacks. It should use encryption, authentication, and other security protocols to ensure data privacy and confidentiality.

Compatibility: The app will be compatible with multiple devices and operating systems, such as Android and iOS, and various screen sizes and resolutions. It should also support multiple languages to cater to a diverse audience.

Scalability: The app will be able to handle increased traffic and user demands during peak hours, such as holidays or weekends, without affecting its performance or functionality. It should be able to scale up or down as per the changing needs of the business.

Accessibility: The app will be accessible to all users, including those with disabilities, through assistive technologies such as screen readers or voice commands. It will also comply with accessibility standards and guidelines to provide an inclusive experience for all users.

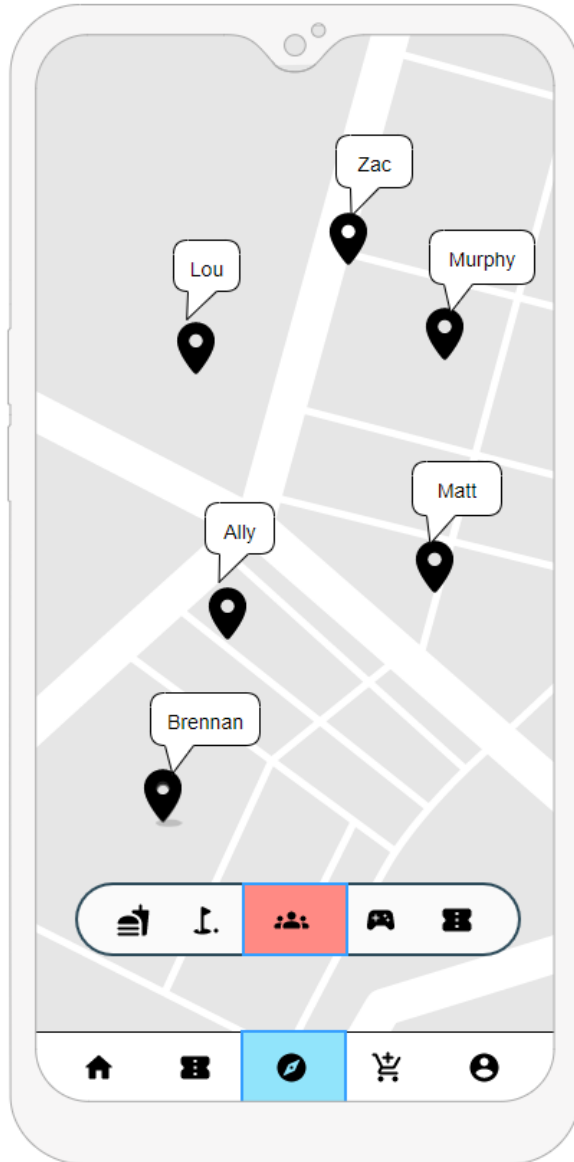
5. References

The References section displays all the references used to make this document. The references will be cited in the main text and then posted down here in the references section.

6. User Interface

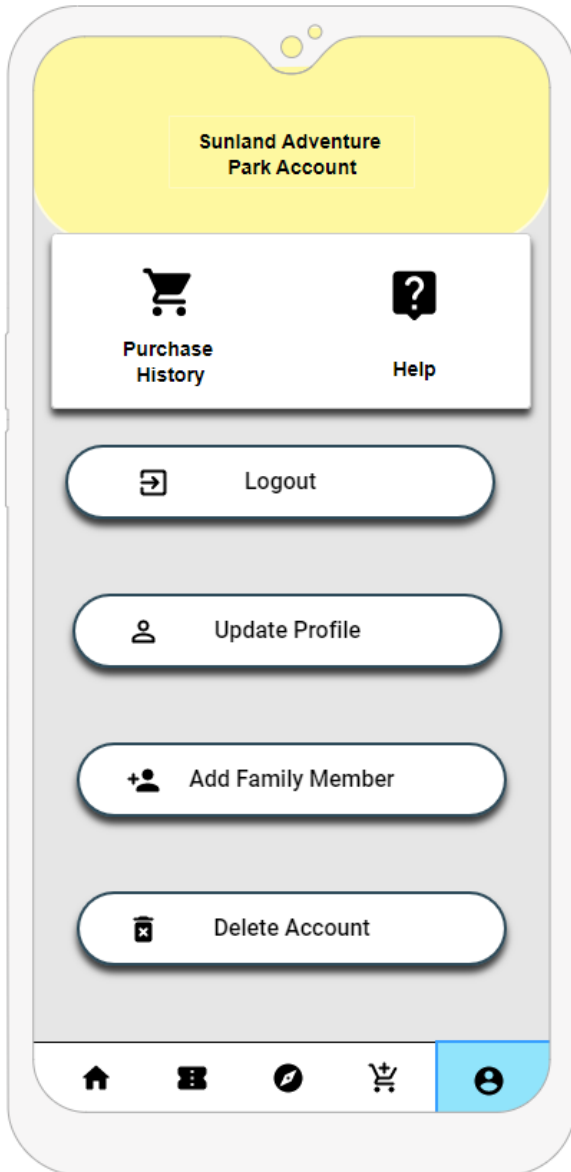
The user interface section displays how the guests interact with the system. We include digital mockups for the app and all of its major pages. The next few pages include the digital mockups for the app.

Figure 6.1 Interactive Map UI



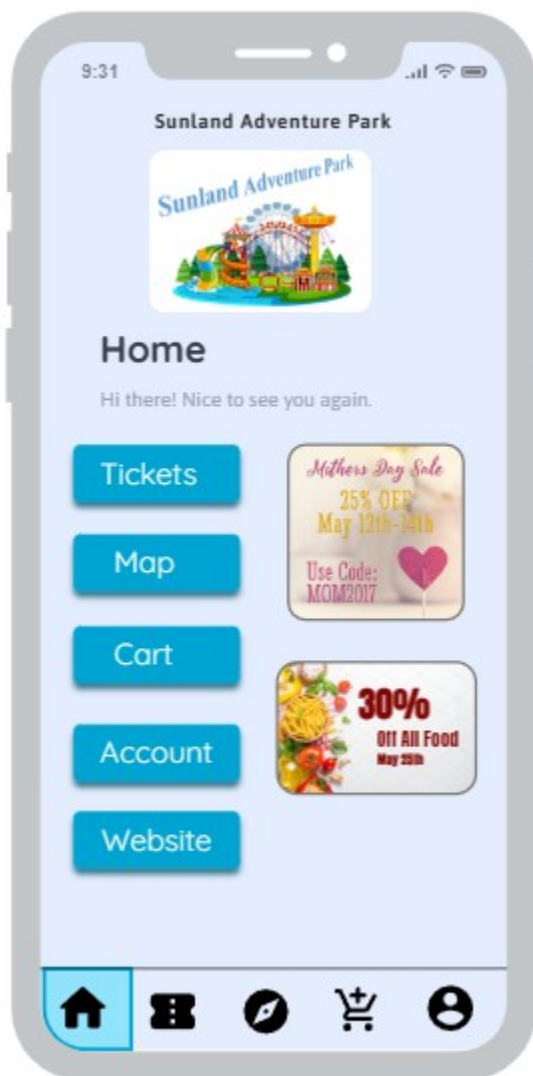
This shows what the UI on the app will look like when a user is looking at the interactive map. The interactive map currently shows friends and their locations but also has options for food, rides, activities, and shows. Separating the options like this ensures the viewer doesn't get distracted by all the different elements displayed.

Figure 6.2 Profile Page UI



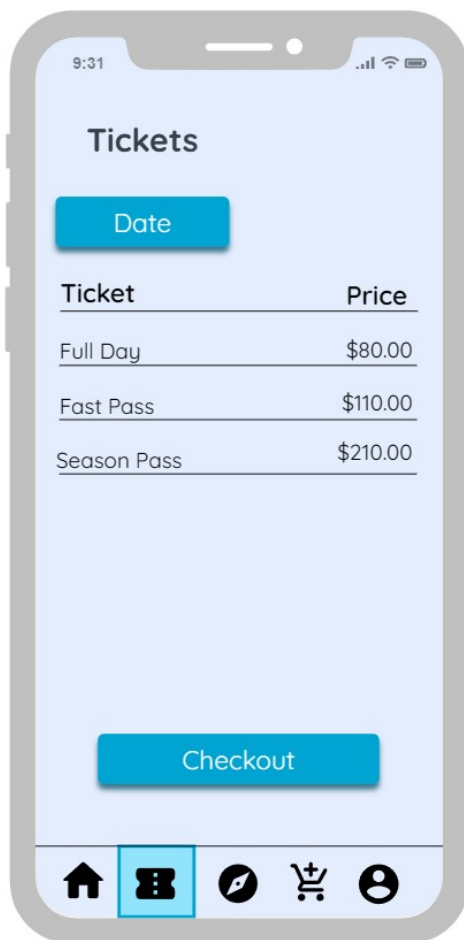
This shows what the UI on the app will look like when a user is looking at things associated with their profile. Guests have options to check their purchase history, ask for help (bugs, FAQ, or general assistance), log out, update their profile, add a family member, and delete an account.

Figure 6.3 Home Page UI



The home page UI shows the home screen for the app. The home screen includes buttons that link to each page. This includes the tickets page, map page, cart page, account page, and a button that brings you directly to the website. The home screen also features the option to run any advertisements or deals the park is currently running.

Figure 6.4 Ticket Page UI



The ticket page UI shows what the ticket page would look like on the app. At the top of the page is a button for the date. Once a guest picks the date they'd like, the app will update with the different kinds of tickets available for purchase. They can then click on the price and add it to their cart. We implemented a checkout button at the bottom so guests have the opportunity to pay immediately through the ticket page. They can also choose to pay through their cart.

7. Appendix

The Appendix section provides the necessary documentation from the references section. The full document from anything recorded in the references section will be included in the appendix section. It is the collection of full documents referenced in the Sunland Adventure Park Software Requirement Specification document.

8. Effort Breakdown Table

Figure 8.1 Effort Breakdown Table:

		Team Member Name:				
Final Project Tasks	points	Ricardo	Vincent	Savhanna	Josiah	N/A
System Request / CBA (120)						
System Request	60	100%	0%	0%	0%	
CBA	60	0%	50%	10%	40%	
Presentation/Brochure (120)						
Presentation	60	0%	50%	50%	0%	
Brochure	60	20%	0%	0%	80%	
SRS (250)						
Customer Statement of Requirements	30	0%	50%	50%	0%	
Glossary of Terms	5	0%	0%	100%	0%	
Stake holders	5	0%	0%	0%	100%	
Actors and Goals	5	0%	0%	0%	100%	
Use case diagram	20	0%	0%	0%	100%	
Casual descriptions	20	0%	0%	0%	100%	
Use case descriptions	20	0%	0%	100%	0%	
Activity diagrams	25	100%	0%	0%	0%	
Interaction diagrams	25	25%	25%	25%	25%	
Overview Class diagram and domain list	25	0%	0%	0%	100%	
Non-funct Requirements	15	0%	100%	0%	0%	
References	5	0%	0%	100%	0%	
User Interface design	50	50%	0%	50%	0%	
Team Member Total		128.25	96.25	112.25	153.25	

The effort breakdown table shows the breakdown of effort among the Group 1 team members. The table itself shows the distribution of work among each member and the percentage amount each member is required to do for each task.